
Estimating Propensity Adjustments for Volunteer Web Surveys

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Richard Valliant¹ and Jill A. Dever²

Abstract

Panels of persons who volunteer to participate in Web surveys are used to make estimates for entire populations, including persons who have no access to the Internet. One method of adjusting a volunteer sample to attempt to make it representative of a larger population involves randomly selecting a reference sample from the larger population. The act of volunteering is treated as a quasi-random process where each person has some probability of volunteering. One option for computing weights for the volunteers is to combine the reference sample and Web volunteers and estimate probabilities of being a Web volunteer via propensity modeling. There are several options for using the estimated propensities to estimate population quantities. Careful analysis to justify these methods is lacking. The goals of this article are (a) to identify the assumptions and techniques of estimation that will lead to correct inference under the quasi-random approach, (b) to explore whether methods used in practice are biased, and (c) to illustrate the performance of some estimators that use estimated propensities. Two of our main findings are (a) that estimators of means based on estimates of propensity models that do *not* use the weights associated with the reference sample are biased even when the probability of volunteering is correctly modeled and (b) if the probability of volunteering is associated with analysis variables collected in the volunteer survey, propensity modeling does not correct bias.

¹University of Michigan and University of Maryland, College Park, MD, USA

²RTI International, Washington, DC, USA

Corresponding Author:

Richard Valliant, University of Maryland, 1218 Lefrak Hall, College Park, MD 20742

Email: rvalliant@survey.umd.edu

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Surveys conducted over the World Wide Web (Web surveys) have several advantages, including shorter data collection periods and lower cost in contrast to alternatives such as telephone and in-person household surveys. Web surveys of persons come in various types. Couper (2000) lists eight designs for Web surveys that range from nonprobability volunteer samples to probability samples from all or part of a target population. In the best circumstances, a frame with e-mail addresses is available for all eligible persons and a sample is asked to visit a Web site and complete a survey. In those cases, the Web serves as just another mode of data collection. In other applications, the Web may also be a vehicle for constructing the frame. A collection of volunteers is recruited by a variety of means (e.g., banner ads on Web pages, e-mail lists, promotions, and offers). Persons who “opt-in” form a pool from which samples are selected for particular surveys. The firm Harris Interactive uses this method for a variety of surveys (Schonlau et al. 2004). Other companies that maintain large online panels include TNS Global, Ipsos MediaCT, Synovate, and YouGovPolimetrix (<http://www.tnsglobal.com>, <http://www.ipsos.com>, <http://synovate.com>, <http://www.polimetrix.com>). Another example is described by Börsch-Supan et al. (2004), who report that the 2001 online survey *Perspektive Deutschland* (www.perspektive-deutschland.de) was conducted with a volunteer sample of more than 170,000 persons. In the 2005-2006 wave, this survey had more than 600,000 volunteers.

To make inferences to a population based on a volunteer sample, some strong assumptions are required; these are described in later sections. To understand the assumptions, it is useful to specify three groups associated with a Web survey:

- (i) *target population*—the set of persons for which inferences are to be made;
- (ii) *potential covered population*—persons having access to the Internet and who use it; and
- (iii) *realized sample*—persons who volunteer to be part of an opt-in panel.

We use “potential” to describe (ii) because this group includes all of the persons who conceivably could volunteer. Volunteer surveys suffer from

at least two defects: (a) The *potential covered population* (ii) may not encompass the full *target population* (i) for which estimates are desired, and (b) self-selection can bias estimates because the *realized sample* (iii) may not be a good representation of either the target population or the potential covered population. Defect (a) means that an important segment of the population—those without Web access or who have access but do not use the Web—is unreachable. Defect (b) implies that the reachable respondents may be a biased sample of the potential covered population. Among persons with Web access, volunteers are unlikely to be distributed by age, gender, and so on, in the same way as either the target population or the potential covered population. Because of this, the volunteers (iii) may over- or underrepresent parts of (i) or (ii). If the potential covered population omits a particular demographic group (e.g., elderly women), then inferences obviously cannot be made for that group unless it behaves in almost the same way as a group in (iii). Inferences for the full populations in either (i) or (ii) may be also biased, with the degree depending on the size of the omitted group and on how much their characteristics differ from those of the rest of the population.

Of course, these general criticisms are not unique to Web surveys but seem more worrisome because of the way the volunteer pool is formed. The same two criticisms may be leveled against landline telephone surveys because coverage there is incomplete and response rates can be low. However, the fact that initial selection can be controlled by the telephone survey designer means that a random sample can be selected from the potential covered population. This contrasts with a Web survey where there is currently no way to identify all Internet users. There is also a large number of persons who would never sign up for an opt-in panel even though they have Internet access. Thus, the gap between the potential and actual covered populations in a volunteer Web survey is much larger than in a telephone survey where (ii) would be the set of persons with telephones and (iii) would be the respondents.

Although coverage errors and low response rates do not necessarily portend biased estimates (see, e.g., Groves 2006), steps are usually taken in weighting to try to correct potential biases. There are several options for how to use a nonprobability sample in estimation. Elliott and Haviland (2007) describe a methodology for combining a convenience or volunteer sample with a probability sample in order to produce an estimator with a smaller mean-squared error than those of estimators based on only the probability sample. This requires that the same analysis variables be collected in both surveys. Another approach, which is the subject of this article, is to devise a set of weights for the volunteer sample to make it representative of the

target population. First, an attempt is made to estimate a probability of participation in the Web survey. This is typically followed by calibration to population counts from a census or a large household survey that, ideally, has minimal coverage problems of the target population. Consider a general case where an opt-in panel of volunteers has been recruited and a probability subsample of the panel is selected and asked to participate in a particular survey. The probability of person k participating in that Web survey can be decomposed as

$$\pi(\mathbf{x}_k) = \pi(W|\mathbf{x}_k)\pi(V|W, \mathbf{x}_k)\pi(s_V|V, W, \mathbf{x}_k)\pi(s_{V_R}|s_V, V, W, \mathbf{x}_k), \quad (1)$$

where

- $\mathbf{x}_k$ = a vector of covariates for person k that are predictive of participation;
- $\pi(W|\mathbf{x}_k)$ = probability of having access to the Internet;
- $\pi(V|W, \mathbf{x}_k)$ = probability of volunteering for an opt-in panel given that person k has access to the Internet;
- $\pi(s_V|V, W, \mathbf{x}_k)$ = probability that person k was subsampled from the panel and asked to participate, with s_V denoting the subsample from the panel;
- $\pi(s_{V_R}|s_V, V, W, \mathbf{x}_k)$ = probability that person k responds, given selection for the subsample, with s_{V_R} denoting the set of survey respondents.

If the product in equation (1) were available for each person in the volunteer sample, then $1/\pi(\mathbf{x}_k)$ could be used to construct a Horvitz-Thompson-like estimator of a population total. The probabilities $\{\pi(\mathbf{x}_k)\}_{k \in s_{V_R}}$ expand the responding volunteers in the subsample from the realized sample (iii) to the target population (i). In this article, we assume that $\pi(s_V|V, W, \mathbf{x}_k)$ and $\pi(s_{V_R}|s_V, V, W, \mathbf{x}_k)$ can be correctly tracked or estimated, respectively, using standard methods. This is not meant to minimize the importance of nonresponse among selected panel members, which may be substantial. Rather, it lets us focus on how a volunteer panel can be used to make inferences for a population. Thus, we concentrate on the case

$$\pi(\mathbf{x}_k) = \pi(W|\mathbf{x}_k)\pi(V|W, \mathbf{x}_k). \quad (2)$$

A key point is that a survey researcher does not control the assignment of $\pi(\mathbf{x}_k)$ as she or he would in a probability sample. The $\pi(\mathbf{x}_k)$ can be considered only as a pseudo-selection probability, which is similar to the thinking used

when computing nonresponse adjustments in surveys. In the nonresponse literature, modeling response as a random process is known as a “quasi-randomization assumption”—a term that we will also use here.

A variation of propensity scoring is used by some organizations to attempt to estimate expression (2). Assumptions about the volunteering mechanism are needed to justify the propensity adjustments, but these have been glossed over or ignored by many authors. Our goals in this article are to clarify the assumptions, to present the mathematics needed to justify the methods used to estimate the propensities, to explain how the assumptions can be violated, and to illustrate how the propensity-based estimates perform in a simulation. In the following, the second section contains examples of current methods implemented to produce population estimates from volunteer Web surveys, including the use of a reference survey that is representative of a defined target population. The third section discusses how the survey population can be segmented based on whether persons are in-scope of the reference and volunteer surveys and relates these to the target and potential covered populations. The fourth section sketches the mechanics behind the creation of propensity scores for use in computing weights for a volunteer survey. The fifth section describes why some propensity-weighted estimators are biased. Through a simulation study described in the sixth section, we demonstrate the empirical properties of the estimators and compare them to a regression estimator that does not use propensities. We summarize our findings in the last section.

Approaches to Estimation

There are at least three avenues for developing estimators for volunteer Web surveys. One is a pseudo-randomization approach where a probability of participation is estimated for each unit, as described above. The inverse of that probability is used as a weight. A second method is to construct estimators based on models, as suggested by Deville (1991) for quota samples. The model can be a simple one where means differ among groups, leading to the poststratified estimator (e.g., see Bethlehem 2010), or a more elaborate one involving multiple covariates, as used by Dever, Rafferty, and Valliant (2008). The third is a combination where, first, the participation probability is estimated and then further adjustments are made based on a structural population model.

There are many studies available, at this point, where one or more of these methods have been applied to Web surveys. Isaksson and Forsman (2003) studied the use of a reference sample in conjunction with a Web sample for estimation. One of their main interests was determining how extensive a set of covariates was needed for estimating propensity models. Lee

(2006) and Lee and Valliant (2009) studied several alternative ways of using propensity scores, including propensity poststratification, which we describe in detail in the section, Mechanics of Estimating Propensity Scores. Schonlau et al. (2004) presented results where poststrata based on propensity scores were combined with other more traditional poststratifiers, like race, age, gender, and income, in a raking estimator. Schonlau, van Soest, and Kapteyn (2007) addressed the question of covariate selection, particularly whether “lifestyle” questions might be useful in estimating propensities of participating in a Web survey. Schonlau et al. (2009) compared persons with and without access to the Internet in the Health and Retirement Survey (HRS). Along with other alternatives, they examined whether a weight equal to the HRS weight divided by an estimated propensity of having the Internet could be used to make estimates for the full population. Earlier research includes work by Terhanian (2000), Terhanian et al. (2000), Varedian and Forsman (2003), and Vehovar, Manfreda, and Batagelj (1999). There is also a literature on inference from nonrandom samples that covers some of the issues that arise with volunteer Web samples (Smith 1983; Copas and Li 1997).

These approaches are heavily burdened with assumptions about the mechanisms that describe access to the Web, willingness to participate in a Web survey, and the superpopulation model(s) for the analysis variables. In the fourth section we examine how propensity scoring has been applied to estimating the participation probability in equation (2).

The method of propensity scoring was developed by Rosenbaum and Rubin (1983) to estimate the unknown probability of a unit being assigned to some treatment in an observational study. Unlike in a randomized trial, units in an observational study are not assigned to treatment or control group using a *known* random mechanism. In that way, volunteer Web samples bear a strong resemblance to observational studies. Units self-select to be in a volunteer survey and, thus, constitute a nonprobability sample. As a result, the usual design-based method of inference found in texts like those by Kish (1965) and Särndal, Swensson, and Wretman (1992) cannot be used.

One approach to inference using volunteer surveys that has been proposed is to apply the Rosenbaum-Rubin method to the set of volunteers (Terhanian 2000; Terhanian et al. 2000; Varedian and Forsman 2003) combined with a reference sample. The reference sample is, ideally, randomly selected from the full target population, that is, the one for which inferences are desired. However, there is an important difference between the goals of estimation in an observational study and in a volunteer survey. In the former, propensity scoring serves to balance or equalize two groups (say, persons exposed and unexposed to a risk factor) on a set of covariates. Given that

balance, the goal is to compare statistics, like the means, of the two groups on some important analytic variables. These means will not be comparable if the groups still differ on key covariates even after the balancing. In fact, uncovering differences between the groups, not explained by differences in the covariate distributions, is the goal of the analysis. On the other hand, in a volunteer survey, the goal is to make estimates that apply to an entire population that includes volunteers and nonvolunteers. The propensity-adjusted volunteer data are used for the estimation; that is, the volunteers must somehow be made to represent the whole population. Comparing the volunteer and reference samples after propensity adjustment is not the goal. If the volunteers differ substantially from the full population, a fundamental question is whether any amount of propensity adjustment will be sufficient to allow inference using the volunteer data.

The target population can be partitioned into subsets covered and not covered by the reference and volunteer surveys. To understand how much of a population we can legitimately make inferences for, we need to understand this partition. This is considered in some depth in the next section.

Universe Coverage

Propensity scores, like other statistical estimates, need to be interpreted as estimating either model parameters or finite population quantities. With either interpretation, it is useful to think about exactly what portions of a finite population are being included or excluded by any procedure proposed for use. In the volunteer–reference case, we can split the target universe (i.e., the population for which inferences are desired) into various subuniverses. Figure 1 breaks the full universe into groups based on whether persons are covered by the reference survey and whether they have Web access. In the figure, $U_{\ell m}$ denotes a subset of the universe (or target population) for $\ell = R, R^c$, or “+” and $m = W, W^c$, or “+”; $s_{\ell V}$ is a subset of the volunteer sample. For example, U_{RW} is the subuniverse of persons who have Web access and who are also covered by the reference survey, s_{RV} is the realized set of volunteers within U_{RW} for the particular volunteer survey, and $s_{RV^c} = U_{RW} - s_{RV}$ is the set of persons in-scope of the reference survey who have Web access but are not volunteers. Relating Figure 1 to the components of the universe introduced in the Introduction, U_{++} is the target population, U_{+W} is the potential covered population, and s_{+V} is the realized volunteer sample. The estimation problem, simply stated, is how to project the data for s_{+V} to U_{++} .

The reference survey itself may have coverage deficiencies. If the reference survey covered landline telephones only, for example, then the part

Coverage of universe by Reference survey	Web available				Total
	Yes			No	
	Volunteer	Non-Volunteer	Total Web		
Covered	s_{RV}	$s_{RV^c} = U_{RW} - s_{RV}$	U_{RW}	U_{RW^c}	U_{R+}
Not covered	s_{R^cV}	$s_{R^cV^c} = U_{R^cW} - s_{R^cV}$	U_{R^cW}	$U_{R^cW^c}$	U_{R^c+}
Total	s_{+V}	$s_{+V^c} = U_{+W} - s_{+V}$	U_{+W}	U_{+W^c}	U_{++}

Figure 1. Breakdown of the target universe into subuniverses based on coverage by the reference or volunteer surveys

Note: A superscript *c* denotes complement; for example, s_{RV^c} is the complement of s_{RV} within the subuniverse with Web access that is covered by the reference survey, U_{RW} .

of the universe not covered (U_{R^c+}) would be persons who have only a wireless phone or who have no phone at all. Such a breakdown must, naturally, refer to a particular point in time because the set of persons having a telephone or access to the Web varies over time. The size of $s_{+V} = s_{RV} \cup s_{R^cV}$, the Web volunteers, can be fairly large but is also likely to be an extremely small portion of either the full universe (U_{++}) or the subuniverse with Web access, $U_{+W} = s_{RV} \cup s_{R^cV} \cup s_{RV^c} \cup s_{R^cV^c}$. Harris Interactive, for example, advertises that it uses a “multimillion member panel” in the United States (<http://www.harrisinteractive.com/uk/MethodsTools/DataCollection/SpecialtyPanels/PanelDevelopment.aspx>), but if, say the panel had 2 million members, this would be only 0.86 percent of the U.S. population aged 18 years and older, which was estimated to be about 232.5 million as of July 1, 2009 (<http://www.census.gov/popest/national/asrh/NC-EST2009-sa.html>).

The interpretation of the headings in the figure can vary among surveys, depending on operational decisions. “Web available” could mean that a person has Web access at home or it could mean access anywhere, including home, work, libraries, or other locations. By “Web volunteer,” we mean that the person has volunteered for a particular Web survey (or, perhaps, to be a member of a panel formed by a particular survey organization). To justify the step of propensity modeling, being a Web volunteer (i.e., being in s_{+V}) or not is treated as a random variable. This is different from saying that being a Web volunteer is a deterministic phenomenon—a person either is a volunteer or is not—with no random choice involved. Treating volunteering as random is similar to the approach of modeling response as random, as presented by Bethlehem (2002) and Kalton and Kasprzyk (1986), or modeling coverage as random (Kim, Li, and Valliant 2007).

The categorization in Figure 1 allows us to make some important points about what is being estimated by the approaches that are being used in practice. Some of the variations on how the (random) reference and volunteer samples might be obtained are as follows:

Reference sample	Volunteer panel formation
(a) Area probability sample	(a) Recruitment via Web page banner ads, e-mail lists, etc.
(b) Random-digit-dialing (RDD) landline only	(b) Area probability sample
(c) RDD landline + wireless phone	(c) RDD landline + wireless phone
	(d) RDD landline

How the reference sample is obtained affects the size of U_{R+} compared with U_{++} . Throughout this article, we assume that the reference sample is obtained via some type of random sampling. In principle, area sampling gives the best coverage of the full universe, U_{++} . However, even well-designed and -executed area probability samples in the United States under-cover certain groups by as much as 25 percent (U.S. Census Bureau, 2002). The usual method of correcting the undercoverage bias is with weight calibration as in the empirical illustration described in the Empirical Illustrations section.

In comparison, basing the reference survey on a random-digit-dialing (RDD) landline sample causes substantial coverage problems; that is, U_{R+}^{rc} will be large. For example, the U.S. National Health Interview Survey (NHIS), in the period January through June 2008, found that 35.7 percent of persons 25 to 29 years old lived in households with only wireless phones; for persons age 45 to 64 and age 65 and older, the percentages were 9.2 and 2.8, respectively (Blumberg and Luke 2008).

The NHIS also found differences in analysis variables between wireless households and households that had landlines. The prevalence of binge drinking (i.e., having five or more alcoholic drinks in one day during the past year) among wireless-only adults (37.7 percent) was more than twice as high as the prevalence among adults living in landline households (17.2 percent). Wireless-only adults were also more likely to be current smokers (30.2 percent vs. 18.4 percent for landline households) and less likely to have health insurance (28.3 percent uninsured vs. 13.6 percent uninsured for landline households).

How the volunteer panel is formed affects the size and composition of s_{+V} and, consequently, s_{+V}^{rc} , the nonvolunteers. If an uncontrolled method of recruitment via banner ads on Web pages and similar devices is used, then the

resulting panel represents only itself in the conventional sense of repeated sampling in which a survey designer controls how units are selected for the sample. Inference from s_{+V} to U_{++} requires model-based estimation. In particular, either the variables measured for the panel members (s_{+V}) must follow the same model as for the nonpanel members ($U_{++} - s_{+V}$) or the act of volunteering can be modeled as a random event (which is the assumption underlying propensity modeling). If the volunteer panel was recruited through an area probability sample, this would ensure that all of U_{+W} had a chance of being covered and offered the opportunity to volunteer. The unwillingness to volunteer could be treated as a type of nonresponse. A panel recruited through an RDD landline and/or wireless phone sample would have the coverage problems noted above.

Mechanics of Estimating Propensity Scores

We turn, now, to how propensities are used in estimation. Suppose that the volunteer sample s_{+V} (referred to as s_V below, for brevity) has n_V persons, and the reference sample is s_R with n_R persons. The combined sample has $n = n_V + n_R$ persons. For the sake of generality, suppose that the units in each survey have “base” weights of some kind associated with them. If the volunteer sample is a subsample of a larger panel, its weights might be the inverses of selection probabilities from the panel. If the reference survey is, for example, a stratified RDD telephone sample, its base (or design) weights could be the inverses of the associated selection probabilities. Here, we denote the base weight of unit i in the volunteer sample by d_{Vi} , where $i \in s_V$; unit j in the Reference survey has base weight d_{Rj} , $j \in s_R$.

The two samples are combined into one, $s_V \cup s_R$, and the estimated propensity of being in the volunteer sample is $\hat{\pi}(\mathbf{x}_k)$. The same set of covariates must be observed for both the volunteer and reference samples to allow a propensity model to be estimated. As noted by Lee and Valliant (2009), the covariates should be measured in the same way in both surveys (e.g., identical question wording). Measurement differences for some covariates like age, sex, and perhaps race may be inconsequential even if different modes are used in the surveys. But, for more complex socioeconomic measures like income and assets, noncomparabilities may be serious. This can lead to biased estimates of propensities.

To estimate the parameters of a propensity model, a choice must be made whether to use the weights that are available for the volunteer and reference surveys. The volunteer weights, if any, inflate the volunteer sample only to the level of the full opt-in panel (s_{+V} in Figure 1). The reference survey

weights can be constructed to inflate the reference sample to either U_{R+} or U_{++} . Strictly speaking, in the former case, the reference survey weights should be adjusted to inflate s_R to $U_{R+} - s_{RV}$, that is, the reference survey universe excluding the part of U_{R+} that consists of volunteers. Typically, s_{RV} is so small relative to U_{R+} that this adjustment is unnecessary. For example, if U_{R+} consists of the approximately 303 million persons in the United States who have either a landline or a cell phone, and the volunteer panel has 1 million persons, all of whom have a telephone, then s_{RV} would be only 0.33 percent of U_{R+} .

If the weights for both the volunteer and the reference surveys are used and the reference survey weights inflate that sample to $U_{R+} - s_{RV}$, the fitted parameters estimate those in the model that would be estimated if all persons in $s_V \cup U_{R+}$ were in the sample. In that case, the estimated propensity for person k estimates $\pi(\mathbf{x}_k) = \pi(W|\mathbf{x}_k)\pi(V|W, \mathbf{x}_k)$ in equation (2) for $k \in s_V \cup U_{R+}$. An alternative is that the reference sample is calibrated to external population counts for U_{++} . In that case, the reference survey weights should be adjusted so that s_R weights up to $U_{++} - s_V$, but this adjustment will also, typically, be negligible. The weighted estimates of propensities will refer to the probability of volunteering within U_{++} in this case.

Apparently, the more common approach is to fit the propensity model using no weights. For example, Schonlau et al. (2004), Isaksson and Forsman (2003), and Börsch-Supan et al. (2004) are silent on whether they used weights in fitting, but the implication is that unweighted regressions were used. If no weights (i.e., all weights equal to 1) are used, then the estimated propensities will inflate the volunteer sample only to the level of the combined sample, $s_V \cup s_R$. Consequently, using only inverse propensity weights from an unweighted regression will result in biased estimates. We cover this in more detail in the section, Properties of the Propensity-Weighted Means.

Three options for using the propensity scores in estimation are listed below. Denote the estimated propensity for person k as $\hat{\pi}(\mathbf{x}_k)$ or in a more compact notation, $\hat{\pi}_k$.

- (a) Use $1/\hat{\pi}_k$ as a weight adjustment for person $k \in s_V$. The estimated mean of an analysis variable Y is $\hat{y}_1 = \sum_{s_V} d_{V_k} y_k / \hat{\pi}_k / \sum_{s_V} d_{V_k} / \hat{\pi}_k$, where y_k is the value observed for volunteer sample unit k and d_{V_k} is the corresponding base weight for unit k .
- (b) Sort the combined sample by $\hat{\pi}_k$; partition the combined sample into G subclasses, each of which has about the same number of cases in the combined sample; and compute an average propensity,

$\hat{\pi}_g$, within subclass g . Use $1/\hat{\pi}_g$ as the weight adjustment for every person in the subclass. The estimated mean is $\hat{y}_2 = \sum_{g=1}^G \sum_{s_{Vg}} d_{Vg} y_k / \hat{\pi}_g / \sum_{g=1}^G \sum_{s_{Vg}} d_{Vg} / \hat{\pi}_g$, where s_{Vg} is the set of volunteer sample units in subclass g .

- (c) Form subclasses as in (b) and use estimated population counts of persons in each subclass to create a type of poststratified estimator.

Methods (a) and (b) are often used for nonresponse adjustments in surveys, with $\hat{\pi}_k$ being the unit's estimated probability of response given that it was selected for the sample. Little (1986) and Czajka et al. (1992) recommend forming classes as in (b) and using a single adjustment for each responder in a class. Using ungrouped estimated propensities can produce highly variable $1/\hat{\pi}_k$ weights, which, in turn, can inflate variances of estimated means. Using a single adjustment in each class can substantially reduce this variability.

One version of (c) is explained in somewhat more detail here since it will be used in the simulation study presented later. Following Cochran (1968), the conventional choice for the number of equal-sized subclasses is five ($G = 5$). Ideally, all units in a given subclass will have about the same propensity score, or at least, the range of scores in each subclass is fairly narrow. If more homogeneous subclasses are desired, more than five subclasses can be used. In the g th subclass in the merged data set, there are $n_g = n_{Vg} + n_{Rg}$ units, where n_{Vg} is the number of sample units from the volunteer survey and n_{Rg} from the reference survey.

The following adjustment factor is calculated and applied to all units in the g th subclass of the volunteer sample:

$$f_g = \frac{\sum_{k \in s_{Rg}} d_{Rk} / \sum_{k \in s_R} d_{Rk}}{\sum_{k \in s_{Vg}} d_{Vg} / \sum_{k \in s_V} d_{Vg}}, \quad (3)$$

where s_{Rg} is the set of units in the reference sample that are in the g th subclass. If the weights in equation (3) are the inverses of some type of selection probabilities, the adjustment is equivalent to $f_g \equiv \frac{\hat{N}_{Rg}/\hat{N}_R}{\hat{N}_{Vg}/\hat{N}_V}$, where $\hat{N}_{\ell g} = \sum_{k \in s_{\ell g}} d_{\ell k}$ is the estimated population count of units in subclass g based on the reference ($\ell = R$) or volunteer ($\ell = V$) survey, and

$\hat{N}_\ell = \sum_g \hat{N}_{\ell g}$ is the corresponding estimated population count. What population count is being estimated by $\hat{N}_\ell$ depends on the construction of the base weights, $d_{\ell k}$, and how these relate to the subuniverses in Figure 1. If d_{Rk} and d_{Vk} are both inverses of selection probabilities, then $\hat{N}_{Rg}$ and $\hat{N}_{Vg}$ are estimates of the population counts of persons on the reference and volunteer survey frames in subclass g (ignoring the randomness of subclass boundaries). With this notation, the adjusted weight for unit k in s_{Vg} is

$$d_k^* = f_g d_{Vk} = \frac{\hat{N}_{Rg} / \hat{N}_R}{\hat{N}_{Vg} / \hat{N}_V} d_{Vk}. \quad (4)$$

Some properties of estimators constructed using these weights are analyzed by Lee and Valliant (2009). Using the weights in equation (4), the estimator for a mean from the volunteer survey sample s_V is

$$\hat{y}_3 = \frac{\sum_g \sum_{k \in s_{Vg}} d_k^* y_k}{\sum_g \sum_{k \in s_{Vg}} d_k^*}. \quad (5)$$

This estimator appears to be the one used by Harris Interactive, although Harris Interactive's method is considered proprietary and its details are not spelled out (see Terhanian and Bremer 2000: 3). One of the properties of this method of weighting is that $\hat{N}_g^* = \sum_{k \in s_{Vg}} d_k^* = \hat{N}_V (\hat{N}_{Rg} / \hat{N}_R)$; that is, the estimated distribution among subclasses of the volunteer sample cases is adjusted to equal the estimated distribution based on the reference sample cases. Because of this, we will refer to $\hat{y}_3$ as a “propensity-poststratified” estimator. Note that this method does not typically equalize the distribution of the volunteer and reference samples on the individual covariates. The factor f_g can also be written as

$$f_g = \frac{1}{\hat{e}_g} \frac{\hat{N}_V}{\hat{N}_R},$$

with $\hat{e}_g = \hat{N}_{Vg} / \hat{N}_{Rg}$ being a type of propensity estimator, assuming that $\hat{N}_{Rg}$, computed using the reference survey weights, is an estimate of the reference population count. Since $\hat{N}_V / \hat{N}_R$ cancels out when computing a mean, equation (5) reduces to $\hat{y}_3 = \frac{\sum_g \sum_{k \in s_{Vg}} d_{Vk} y_k / \hat{e}_g}{\sum_g \sum_{k \in s_{Vg}} d_{Vk} / \hat{e}_g}$, with $d_{Vk} / \hat{e}_g$ being a propensity-adjusted weight.

If all the volunteer base weights are initially 1, then the three estimated means reduce to

$$\begin{aligned}\hat{y}_1 &= \left[\sum_{s_V} y_k / \hat{\pi}_k \right] / \left[\sum_{s_V} 1 / \hat{\pi}_k \right], \\ \hat{y}_2 &= \sum_{g=1}^G \hat{p}_{Vg} \hat{y}_{Vg}, \text{ and} \\ \hat{y}_3 &= \sum_g \hat{p}_{Rg} \hat{y}_{Vg},\end{aligned}\tag{6}$$

where $\hat{p}_{Vg} = (n_{Vg} / \hat{\pi}_g) / \sum_{g=1}^G n_{Vg} / \hat{\pi}_g$, $\hat{y}_{Vg} = \sum_{s_{Vg}} y_k / n_{Vg}$ and $\hat{p}_{Rg} = \hat{N}_{Rg} / \hat{N}_R$, because $n_{Vg} \equiv \hat{N}_{Vg}$ when $d_{Vg} \equiv 1$. If the reference survey base weights are not used or if the reference survey is a simple random sample, then $\hat{p}_{Rg} = n_{Rg} / n_R$, which was the application used by Isaksson and Forsman (2003). As is apparent from the formulas in equation (6), $\hat{y}_2$ and $\hat{y}_3$ differ because of the way in which they combine the unweighted subclass means of the volunteer sample. We will examine the properties of these estimators in the next section.

There is another way to use propensities in estimation, which will not be covered here but is important and deserves mention. After estimating the propensity of volunteering for each unit, a volunteer can then be matched to a unit in the reference survey that has a similar size propensity. The volunteer unit then receives the weight of the unit in the reference survey. Theory for propensity matching was given by Rosenbaum and Rubin (1983). The firm YouGovPolimetrix uses a variation of this method for estimation in its opt-in panel (Rivers, n.d.).

Properties of the Propensity-Weighted Means

The properties of the three estimators of means in the previous section depend on whether the volunteer and reference survey weights are used to estimate propensity scores, as shown heuristically in this section and in more detail in the online appendix (found at <http://smr.sagepub.com/supplemental>). In particular, $\hat{y}_1$ and $\hat{y}_2$ are unbiased when survey weights are used to estimate propensities but biased when weights are not used. Consider the case where s_V is the entire panel of volunteers so that the base weights in the volunteer sample are all 1 ($d_{Vg} \equiv 1$). An important point is that persons who are in the volunteer sample cannot be in the reference sample for the propensity model to be estimated cleanly. This is similar to the situation in an observational study where the treated and the nontreated cases are disjoint. In the literature on volunteer–reference surveys, this is a necessary

but unspoken assumption when propensity modeling is done. Note, however, that there will be overlap between the volunteer subuniverse and the reference universe; that is, s_{RV} in Figure 1 is not empty. For example, if U_{R+} consists of persons with landline phones, there will be some who volunteer to be part of a volunteer panel. As noted in the previous section, weights for the reference survey can be adjusted to weight the sample to $U_{R+} - s_{RV}$ or $U_{++} - s_V$ if such an adjustment is needed.

As shown in the online appendix (found at <http://smr.sagepub.com/supplemental>), when the propensity model is fitted using the weights for the reference survey that inflate it to $U_{R+} - s_{RV}$, $E(\hat{y}_1) \doteq \bar{Y}_{s_V \cup U_{R+}}$, where the average is taken over the quasi-randomization volunteering mechanism and where $\bar{Y}_{s_V \cup U_{R+}}$ is the mean for all units in the combined population, $s_V \cup U_{R+}$, covered by the volunteer and reference samples. If $\bar{Y}_{s_V \cup U_{R+}} = \bar{Y}_{U_{++}}$, the mean for the entire target population, then $\hat{y}_1$ is approximately unbiased for the desired population mean. In contrast, when the propensity model is fitted *without* using weights, then the estimated propensities inflate s_V only to the level of the combined sample, $s_V \cup s_R$. In that case, the conditional expectation with respect to the volunteering mechanism of $\hat{y}_1$, given the combined sample, is just the combined sample mean, $\bar{y}_s = \sum_{s_V \cup s_R} y_k / n$, which can be written as

$$\bar{y}_s = \alpha_V \bar{y}_V + (1 - \alpha_V) \bar{y}_R, \quad (7)$$

where $\alpha_V = n_V / n$, $n = n_V + n_R$, and $\bar{y}_V$ and $\bar{y}_R$ are the unweighted means for the volunteer and reference samples, respectively. Note that this is a theoretical expectation since we do not collect y_s in the reference sample. From the conditional mean in equation (7), it is apparent that even if the mean of the reference sample is an unbiased estimator of the population mean, combining it with a biased volunteer sample estimator leads to an overall bias. Increasing the volunteer sample size does not necessarily reduce the bias if $\bar{y}_V$ does not approach the population mean as the sample size increases. In fact, increasing the volunteer sample size while keeping the reference sample size constant will increase, not decrease, the bias.

To carry the analysis further, a design for the reference sample must be specified. Assuming the reference survey is a simple random sample and averaging over the quasi-random mechanism for being a volunteer, the unconditional bias of $\hat{y}_1$ relative to $\bar{Y}_{s_V \cup U_{R+}}$ is, as shown in the online appendix (found at <http://smr.sagepub.com/supplemental>),

$$relbias(\hat{y}_1) = \alpha_V relbias_I(\hat{y}_V), \quad (8)$$

where $relbias_I(\hat{y}_V) = [E_I(\bar{y}_V)/\bar{Y}_{s_V \cup U_{R+}}] - 1$ and E_I is an expectation over the volunteering mechanism. There are several consequences of equation (8) for unweighted estimation of the propensity scores. As above, assume that $\bar{Y}_{s_V \cup U_{R+}} = \bar{Y}_{U_{+}}$.

- (a) If the unweighted mean of the volunteer sample, $\hat{y}_V$, is a quasi-randomization unbiased estimator of the target population mean, then $\hat{y}_1$ is unbiased. This condition is unlikely to hold.
- (b) If the volunteer sample is substantially larger than the reference sample ($\alpha_V \rightarrow 1$), the empirical relative bias (*relbias*) approaches the *relbias* of $\hat{y}_V$.
- (c) If the volunteer sample is much smaller than the reference sample ($\alpha_V \rightarrow 0$), the *relbias* approaches 0.

As shown in the online appendix (found at <http://smr.sagepub.com/supplemental>), $\hat{y}_2$ is approximately quasi-randomization unbiased when the reference survey weights are used to estimate $\hat{\pi}_g$ as long as the volunteering propensities within a subclass are nearly constant. When an unweighted propensity model is fit, $\hat{y}_2$ is, in general, biased with $bias(\hat{y}_2) \doteq \sum_g P_g \bar{Y}_g [\alpha_{Vg} relbias_I(\hat{y}_g^V)]$, where $\alpha_{Vg} = n_{Vg}/n_V$, and $\bar{Y}_g$ is the population mean for persons in subclass g . Thus, in order for $\hat{y}_2$ to be unbiased, the unweighted mean of the volunteers in each subclass must be an unbiased estimator of the population subclass mean.

The bias of $\hat{y}_3$ appears to be difficult to analyze because the properties of $\hat{p}_{Rg}$ depend on the sample-dependent groups. As shown in expression (6), $\hat{y}_2$ and $\hat{y}_3$ both can be written as weighted combinations of the unweighted volunteer means, $\hat{y}_{Vg}$, in each subclass. For $\hat{y}_2$, the subclass weights are equal to $\hat{p}_{Vg}$ from the volunteer survey, while for $\hat{y}_3$, these weights depend on $\hat{p}_{Rg}$ for the reference survey. As illustrated in the next section, these subclass weights can be substantially different and result in large biases for the propensity-poststratified estimator.

Empirical Illustrations

To support the theoretical findings in the previous sections, we conducted a simulation study using data from the 2003 Michigan Behavioral Risk Factor Surveillance Survey (<http://www.cdc.gov/BRFSS>) as the target population of interest. The survey included a module on Internet usage that asked

Table 1. Covariates, Health Variables, and Internet-Usage Variables Used in the Simulation Study

Type of variable	Description
Covariates	
Age category	1: 18-24 2: 25-34 3: 35-44 4: 45-54 5: 55-64 6: 65 or older
Race category	1: White 2: Black 3: Other
Gender	1: Male 2: Female
Wireless phone	1: Has wireless phone 2: No wireless phone
Education category	1: Did not graduate high school 2: Graduated high school 3: Attended college or technical school 4: Graduated from college or technical school
Income category	1: Less than \$15,000 2: \$15,000 to less than \$25,000 3: \$25,000 to less than \$35,000 4: \$35,000 to less than \$50,000 5: \$50,000 or more
Health-related variables	
Smoke100	Smoked 100 or more cigarettes in lifetime 1 = Yes, 2 = No
Body mass index category	1: Neither overweight nor obese (BMI < 25) 2: Overweight ($25 \leq \text{BMI} < 30$) 3: Obese ($30 \leq \text{BMI}$)
General health (self-reported)	1: Excellent 2: Very good 3: Good 4: Fair 5: Poor
Physical activity	In last month participated in activities such as running, calisthenics, golf, gardening, or walking for exercise 1 = Yes, 2 = No
High blood pressure	Have you ever been told by a doctor, nurse, or other health professional that you have high blood pressure? 1 = Yes, 2 = No

continued

Table 1. continued

Type of variable	Description
Asthma	Have you ever been told by a doctor, nurse, or other health professional that you had asthma? 1 = Yes, 2 = No
Diabetes	Have you ever been told by a doctor, nurse, or other health professional that you had diabetes? 1 = Yes, 2 = No
Cholesterol check	Blood cholesterol is a fatty substance found in the blood. Have you ever had your blood cholesterol checked? 1 = Yes, 2 = No
Binge drinking	At risk for binge drinking based on alcohol consumption responses 1 = Yes, 2 = No
Arthritis	Have been told by a doctor or other health professional that you have some form of arthritis, rheumatoid arthritis, gout, lupus, or fibromyalgia, or have joint symptoms of arthritis. 1 = Yes, 2 = No

whether persons had access to the Internet at home. The various covariates and health-related variables used in this study are shown in Table 1. After dropping cases with either a missing, a “don’t know,” or a “refused” response to any of these variables, 2,845 persons were available for the simulation study. This set was expanded to $N = 50,000$ cases by randomly selecting with replacement 47,155 additional cases from the 2,845. We did this expansion to create a large population from which to select samples with a small sampling fraction.¹

Volunteer and reference samples were repeatedly selected from the expanded finite population of 50,000 cases. Each volunteer sample of size n_V was selected using Poisson sampling with probabilities assigned as described below. We treat a realized volunteer sample as being the full panel in a given simulation run. Thus, every volunteer sample unit has a weight of 1. Each reference sample was selected via simple random sampling without replacement from the persons not selected for the volunteer sample, that is, $N - n_V$. With this plan, the weight for each unit in the reference sample is the same, $N/n_R \doteq (N - n_V)/n_R$. (This approximation is valid since the

Table 2. Parameters Used in Logistic Regression for Propensity Model

Parameter	Value
Mechanism (i)	
Intercept	−0.2
Age categories	−0.6, −0.7, −0.75, −0.8, −1.5, 0
Race categories	−0.8, −0.3, 0
Gender categories	−0.2, 0
Wireless phone	−0.7, 0
Education category	0.1, 0.6, 1.0, 0
Income category	0.2, 0.4, 0.5, 0.6, 0
Mechanism (ii): the above covariates and parameter values plus	
High blood pressure	0.5
Physical activity	−0.5

volunteer sample size is at most 2 percent of the full population in our simulations.)

We used two mechanisms for generating the volunteer sample:

- (i) Probabilities of volunteering depend only on covariates known for each sample unit where the population total for each covariate is known.
- (ii) Probabilities of volunteering depend on the same covariates plus some of the analysis variables collected in the survey.

To assign a probability of volunteering to each person in the population of 50,000 with mechanism (i), we fit a logistic regression model to the expanded population to predict whether a person had Internet access at home based on the categorical variables age, race, gender, wireless phone, education, and income class. The categories are shown in Table 1. Rounded parameter estimates listed in Table 2 were used to generate a probability of volunteering for each unit in the population. These probabilities were scaled to obtain a desired, expected number of persons in a volunteer sample.² Of course, this model does not capture the unknown motivation of a person to volunteer given that the person has Web access at home. The model does, however, allow us to generate volunteer samples whose distribution is considerably different from the general population and reflective of Internet penetration.

For volunteering mechanism (ii), a logistic model was used with the same covariates as in (i) along with two additional variables—whether a person reported being diagnosed with high blood pressure or having engaged in

any physical activity in the last 30 days. Mechanism (ii) is a case where the probability of volunteering cannot be modeled correctly since the analytic variables are unknown for the reference sample cases.

Several alternatives were used to estimate the proportions or means for the analytic variables:

- (1) individual propensity weights in $\hat{y}_1$ (labeled “1: propensity wts” in the tables);
- (2) average propensity weights in each of five subclasses for $\hat{y}_2$ (“2: avg propensity wts”);
- (3) propensity-poststratified estimator, $\hat{y}_3$ (“3: propensity PS”); and
- (4) calibration to population totals of covariates in Table 1 (no propensity adjustment) using a regression estimator. The estimator in this study was a type of best linear unbiased predictor (Valliant, Dorfman, and Royall 2000), which is also a general regression estimator (GREG) with all weights equal to 1, as described by Särndal et al. (1992). This choice is labeled “4: calibration to X” in the tables and $\hat{y}_{cal}$ in the text.

GREG estimation was also applied in work by Dever et al. (2008) and Lee and Valliant (2009), where more description of its properties can be found. We have included $\hat{y}_{cal}$ here because it is a standard way of attempting to weight a sample to a population using covariates whose population totals are available. Calibration estimation also does not require a reference survey and, consequently, is a cheaper alternative. Choices (1) to (3) were computed from propensity models fitted with and without reference survey weights. This gives a total of seven alternative estimators. Note that another choice would be a combination of propensity estimation followed by calibration. We omitted this choice so that the effects of the propensity adjustment and calibration are clearly separated. The following combinations of reference sample size and expected value of the volunteer sample size were used:

Volunteer	Reference
250	1,000
500	500
1,000	250

Each combination of sample size and volunteering mechanism was repeated 10,000 times in a simulation program written in **R** and its survey package (R Development Core Team 2009; Lumley 2010). For each combination

of sample size, estimation strategy, covariate, and analysis variable, we computed the empirical relative bias (*relbias*) across the 10,000 samples. Each *relbias* was calculated as $relbias(\hat{\theta}) = 100(\bar{\hat{\theta}} - \theta) / \theta$, where θ is the finite population value from the simulated population of size 50,000; and $\bar{\hat{\theta}}$ is the average estimate across the 10,000 samples. Note that $relbias(\hat{\theta}) = 0$ would be ideal. We summarize some of the results below.

Based on the theoretical results in the previous section and the online appendix (found at <http://smr.sagepub.com/supplemental>), we anticipate that $\hat{y}_1$ and $\hat{y}_2$ will be approximately unbiased if the reference survey weights are used to estimate propensities. If the reference survey weights are not used and unweighted estimates of propensities are used, $\hat{y}_1$ and $\hat{y}_2$ are expected to be biased as long as the volunteer and reference survey means are different—see equations (7) and (8). Moreover, the bias of $\hat{y}_1$ and $\hat{y}_2$ should increase as the volunteer sample size increases from 250 to 1,000 when unweighted estimates of propensity model parameters are used.

Table 3 gives some measures of how balanced the reference and volunteer samples are on the distribution of the covariates and how well the propensity weighting alternatives correct any imbalances. The volunteer samples have *relbiases* that reflect the fact that the population with Web access at home is younger and better educated and has higher income than the population as a whole (see the “Unweighted means” columns in Table 3). Volunteers are also more likely to be White, male, and wireless phone users. The reference samples are balanced on the covariates because they are simple random samples from the nonvolunteers, and the volunteers are a small part of the full population. With unweighted parameter estimates, none of the propensity weighting methods correct the imbalance in the volunteer samples. In fact, propensity poststratification (3: propensity PS) has *relbiases* similar to those of the volunteer samples themselves. On the other hand, weighted estimates of the parameters in the propensity models yield smaller *relbiases* for $\hat{y}_1$ and $\hat{y}_2$ (1: propensity wts and 2: avg propensity wts). In contrast, the propensity poststratification estimators based on the weighted parameter estimates are generally no better than the unweighted.

Figure 2 is a dot plot (Wickham 2009) of the *relbiases* for the categorical health variables in Table 1 for sample sizes of 500 in each of the reference and volunteer samples. The left-hand panel shows propensity-adjusted estimates where the propensity model was estimated without using weights for the reference survey; the right-hand panel shows the same estimates when reference survey weights are used. Both panels also show *relbiases*

Table 3. Percentage Relative Biases of Estimated Covariate Totals in 10,000 Samples for the Three Estimation Methods Using Unweighted Means From the Volunteer and Reference Samples and Propensities (1: propensity wts, 2: avg propensity wts, 3: propensity PS) for Reference and Volunteer Sample Sizes of 500

	Propensity estimates							
	Unweighted means		Unweighted parameter estimates				Weighted parameter estimates	
			$\hat{y}_1, 1:$ propensity wts	$\hat{y}_2, 2:$ avg propensity wts	$\hat{y}_3, 3:$ propensity PS	$\hat{y}_1, 1:$ propensity wts	$\hat{y}_2, 2:$ avg propensity wts	$\hat{y}_3, 3:$ propensity PS
Volunteer	Reference							
Age								
18-24	41.5	-0.3	21.9	24.4	37.6	3.3	8.4	37.9
25-34	20.0	-0.3	10.4	11.2	17.9	1.5	3.3	18.0
35-44	16.9	0.0	8.9	9.6	15.3	1.1	2.9	15.5
45-54	10.1	-0.1	5.4	6.4	9.3	0.6	2.8	9.4
55-64	0.6	-0.1	0.3	1.3	0.7	0.3	2.2	0.5
65+	-49.8	0.5	-26.2	-29.8	-45.2	-3.6	-11.6	-45.5
Race								
White	4.1	0.0	2.2	2.5	3.7	0.4	1.0	3.7
Black	-38.5	0.3	-20.9	-23.8	-35.0	-3.5	-10.2	-35.0
Other	-7.2	0.0	-3.9	-4.0	-6.4	-0.5	-0.9	-6.4
Gender								
Male	5.0	0.0	2.8	3.5	4.6	0.7	2.1	4.8
Female	-3.6	0.0	-2.0	-2.5	-3.3	-0.5	-1.5	-3.4
Wireless phone								
Yes	31.7	-0.3	16.5	17.4	28.4	1.6	3.7	28.5
No	-30.2	0.3	-15.7	-16.6	-27.0	-1.5	-3.5	-27.1

continued

Table 3. continued

	Unweighted means	Propensity estimates						
		Unweighted parameter estimates			Weighted parameter estimates			
		$\hat{y}_1, 1:$	$\hat{y}_2, 2: \text{avg}$	$\hat{y}_3, 3:$	$\hat{y}_1, 1:$	$\hat{y}_2, 2: \text{avg}$	$\hat{y}_3, 3:$	
Volunteer Reference propensity wts propensity PS propensity wts propensity PS propensity wts propensity PS								
Education								
Did not graduate high school	-50.0	0.2	-26.6	-31.3	-45.7	-3.0	-13.8	-45.8
Graduated high school	-34.4	0.2	-17.8	-18.3	-30.6	-1.6	-3.1	-30.8
Attended college or technical school	6.0	0.1	3.4	4.3	5.6	0.6	2.8	5.7
Graduated from college or technical school	42.7	-0.4	22.1	23.1	38.0	2.0	4.5	38.1
Income								
Less than \$15,000	-48.1	0.4	-25.2	-28.7	-43.6	-2.8	-10.8	-43.7
\$15,000 to less than \$25,000	-27.2	0.2	-14.0	-13.7	-23.9	-0.8	-0.8	-24.0
\$25,000 to less than \$35,000	4.8	0.1	2.8	3.9	4.6	0.6	3.2	4.6
\$35,000 to less than \$50,000	22.6	-0.5	11.7	12.7	20.3	1.4	4.0	20.4
\$50,000 or more	44.3	-0.3	22.8	23.5	39.4	1.4	3.3	39.5

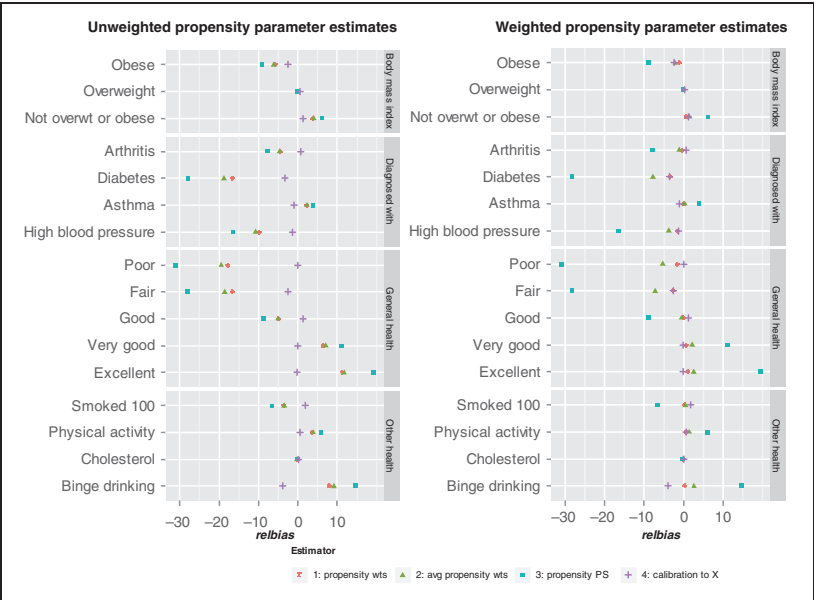


Figure 2. Percentage relative biases in 10,000 samples of estimates of health characteristics with the four alternative methods of estimation
Note: Sample sizes are 500 each for the reference and the volunteer samples; the volunteering mechanism (i) depends only on known covariates.

of the GREG estimates, which are not affected by how the propensity model is estimated. Calibration to the covariates in Table 1 with the GREG $\hat{y}_{cal}$ (“4: calibration to X”) gives nearly unbiased estimates for every health variable. The first two choices of propensity-based estimates, $\hat{y}_1$ and $\hat{y}_2$, are quite biased when reference weights are not used to fit the model, but the bias reduces substantially when weights are used (as predicted theoretically in the previous section). For example, the biases for $\hat{y}_1$ and $\hat{y}_2$ using the poor general health variable are -16.6 percent and -18.6 percent in the left-hand panel but are -2.6 percent and -7.2 percent in the right-hand panel. The performance of $\hat{y}_2$ could probably be improved in the weighted case by using more than five classes, leading to propensity adjustments that are closer to those of $\hat{y}_1$. The propensity-poststratified estimator, $\hat{y}_3$ (“3: propensity PS”), is the worst performer, having biases beyond -27 percent for diabetes and poor or fair general health in the left-hand panel; there is almost no change in these in the right-hand panel.

The difference between $\hat{y}_2$ and $\hat{y}_3$ is interesting since both are combinations of the same unweighted volunteer sample means for each propensity subclass. The factors used to combine these means are $\hat{p}_{Vg} = \hat{N}_{Vg} / \sum_{g=1}^G \hat{N}_{Vg}$ with $\hat{N}_{Vg} = n_{Vg} / \hat{\pi}_g$ for $\hat{y}_2$, and $\hat{p}_{Rg} = n_{Rg} / n_R$ for $\hat{y}_3$. The means of these estimated proportions over the simulated samples are shown below. The subclasses are ordered from low to high in terms of the probability of volunteering. The average estimated proportions in the subclasses are much different for reference and volunteer samples. The reference proportions are close to uniform, that is, near 0.2 in all subclasses, while the estimates from the volunteer samples have much larger shares in the lower probability subclasses. As observed in the online appendix (found at <http://smr.sagepub.com/supplemental>), $\hat{p}_{Vg}$ used in $\hat{y}_2$ will be approximately unbiased for the population proportion in class g in the weighted propensity models as long as the propensity model is specified correctly. As a result, $\hat{p}_{Rg} = n_{Rg} / n_R$ must be biased since it has a much different form than $\hat{p}_{Vg}$ and much different empirical averages. This leads to the substantial biases observed for $\hat{y}_3$. Consequently, although the mathematics needed to anticipate the large biases in $\hat{y}_3$ is not obvious, the message from the simulations is clear—propensity poststratification should not be used.

	Class					Total
	1	2	3	4	5	
Reference samples $\hat{p}_{Rg}^a$	0.212	0.201	0.198	0.198	0.191	1.000
Volunteer samples $\hat{p}_{Vg}$						
Unweighted propensity models ^b	0.313	0.212	0.178	0.159	0.139	1.000
Weighted propensity models ^b	0.412	0.225	0.160	0.120	0.082	1.000

a. Mean of 60,000 samples with reference sample sizes of 250, 500, and 1,000—the number of samples corresponds to the total number of volunteer samples generated.
b. Means over 30,000 samples of volunteer sample sizes of 250, 500, and 1,000.

Table 4 shows the effects of different volunteer sample sizes on $\hat{y}_1$, the estimator based on individual propensity estimates. The *relbiases* for $\hat{y}_1$ (“1: propensity wts”) are given for the three combinations of sample sizes. With mechanism (i)—that is, volunteering dependent only on covariates—*relbiases* are often large when parameter estimates from unweighted propensity models are used. Note that there is a consistent gradation of the size of *relbias* depending on the size of the volunteer sample—the larger the volunteer sample, the larger the *relbias* (see the “Unweighted propensity

Table 4. Percentage Relative Biases in 10,000 Samples for Individual Unit Propensity Weighting $\hat{y}_i$ (1: propensity wts) for Three Combinations of Reference and Volunteer Sample Sizes

	Unweighted propensity parameter estimates			Weighted propensity parameter estimates		
	Sample sizes			Sample sizes		
Reference	250	500	1,000	250	500	1,000
Volunteer	1,000	500	250	1,000	500	250
Body mass index						
Not overweight or obese	5.6	3.7	2.0	0.5	0.7	0.8
Overweight	-0.2	0.1	0.0	0.0	0.1	0.1
Obese	-8.0	-5.7	-3.0	-0.8	-1.1	-1.3
General health						
Excellent	17.3	11.2	5.5	0.4	1.1	1.2
Very good	9.8	6.4	3.2	0.3	0.7	0.9
Good	-7.9	-5.0	-2.2	0.2	-0.2	-0.3
Fair	-25.1	-16.6	-8.7	-1.8	-2.6	-3.2
Poor	-27.7	-17.8	-9.0	-1.0	-1.6	-1.4
Diagnosed with						
High blood pressure	-14.5	-9.8	-5.1	-1.3	-1.5	-1.8
Asthma	3.6	2.2	1.0	0.0	0.0	0.3
Diabetes	-25.2	-16.7	-8.7	-2.4	-3.3	-3.4
Arthritis	-6.9	-4.4	-2.0	-0.1	-0.4	-0.5
Other health measures						
Binge drinking	12.8	8.0	3.6	-0.2	0.2	0.6
Cholesterol check	-0.2	-0.1	-0.1	-0.1	-0.1	-0.1
Physical activity in last 30 days	5.3	3.5	1.8	0.3	0.6	0.7
Smoked 100	-6.1	-3.6	-1.4	0.5	0.3	0.2

parameter estimates” columns in Table 4). This is consistent with the theoretical result (8) in the previous section. When the propensity model is fitted using reference survey weights (columns labeled “Weighted propensity parameter estimates”), the *relbiases* are near 0 for most estimates. The *relbiases* do generally decrease as the volunteer sample sizes increase. The volunteer sizes of 250, 500, and 1,000 correspond to average probabilities of volunteering of 0.005 (250/50,000), 0.01, and 0.02. Thus, an increase in the volunteer sample size corresponds to an increase in the probability of volunteering. Determining whether this is related to the pattern of *relbiases* would take more study.

Table 5. Percentage Relative Biases in 10,000 Samples for General Regression Estimation $\hat{y}_{cal}$ (4: calibration to X) for Volunteer Sample Sizes of 250, 500, and 1,000

	Volunteer sample size		
	250	500	1,000
Body mass index			
Not overweight or obese	1.3	1.3	1.3
Overweight	0.3	0.5	0.3
Obese	-2.5	-2.6	-2.4
General health			
Excellent	0.0	-0.2	-0.3
Very good	-0.1	-0.1	-0.1
Good	1.1	1.2	1.3
Fair	-2.8	-2.5	-2.6
Poor	-0.4	0.0	-0.2
Diagnosed with			
High blood pressure	-1.6	-1.4	-1.3
Asthma	-1.0	-1.0	-0.9
Diabetes	-3.5	-3.4	-3.7
Arthritis	0.6	0.7	0.7
Other health measures			
Binge drinking	-3.8	-3.9	-3.8
Cholesterol check	0.0	0.1	0.1
Physical activity in last 30 days	0.5	0.6	0.6
Smoked 100	1.6	1.8	1.7

Table 5 gives *relbiases* of the GREG for volunteer sample sizes of 250, 500, and 1,000. These are generally small although often somewhat larger than the ones for the weighted propensity estimator $\hat{y}_1$ (1: propensity wts) in Table 4. Table 6 shows *relbiases* for the three propensity estimators based on weighted estimates of model parameters along with those for the GREG in a case where the probability of volunteering depends on covariates and analysis variables. When volunteering depends on Y s—mechanism (ii) above—some estimates have substantial biases. As might be expected, no estimator performs consistently well in this situation. For example, the *relbiases* for the estimates for high blood pressure and physical activity, which are in the model for volunteering, are 20.3 percent and -7.3 percent, respectively, for $\hat{y}_1$ (1: propensity wts). Other analysis variables that were not in the volunteering model, like poor general health and diabetes, are substantially biased for all choices of estimator.

Table 6. Percentage Relative Biases in 10,000 Samples for Propensity Weighted Estimates and the General Regression Estimation (GREG) for Volunteer and Reference Sample Sizes of 500

	Weighted parameter estimates			GREG
	$\hat{y}_1, 1:$ propensity wts	$\hat{y}_2, 2:$ avg propensity wts	$\hat{y}_3, 3:$ propensity PS	$\hat{y}_{cal}, 4:$ calibration to X
Body mass index				
Not overweight or obese	-3.1	-2.4	1.8	-2.4
Overweight	0.0	0.0	0.1	0.3
Obese	4.6	3.6	-2.8	3.2
General health				
Excellent	-5.8	-4.3	12.2	-6.3
Very good	-2.8	-1.3	7.3	-3.2
Good	2.1	1.8	-5.8	3.0
Fair	4.9	0.4	-20.2	4.1
Poor	19.0	15.0	-14.0	20.1
Diagnosed with				
High blood pressure	20.3	17.9	4.6	20.0
Asthma	1.5	1.7	5.7	0.8
Diabetes	9.0	4.3	-16.7	8.2
Arthritis				
Other health measures	1.3	0.5	-5.2	1.8
Binge drinking	-1.9	0.4	11.3	-4.5
Cholesterol check	0.6	0.5	0.7	0.7
Physical activity in last 30 days	-7.3	-6.4	-0.5	-7.0
Smoked 100	0.8	0.9	-5.8	1.7

Note: Probability of volunteering depends on covariates and analysis variables.

Conclusion

Surveys are being done over the Web using sets of volunteers rather than probability samples. Adjusting results from a survey of volunteers using covariates collected in a more well-controlled reference survey is an attempt to correct biases in the volunteer sample. One approach to analyzing estimators is to model the act of volunteering as a quasi-random process where every person in the population has some probability of being a volunteer. We investigated several alternatives that have been proposed for making estimates from the volunteer sample data based on estimating propensities of

volunteering. Key requirements for quasi-randomization inference based on a sample of volunteers are as follows:

- The propensity of volunteering can be modeled using covariates available on both the volunteer and reference sample persons.
- The probability of volunteering cannot depend on any analysis variables collected only in the volunteer sample.
- The propensity model should be estimated in such a way that the estimates apply to the universe represented by the combination of the volunteer and reference samples.

To meet the third requirement, it is critical to use the survey weights associated with the reference survey to estimate the parameters of the propensity model in order to obtain approximately unbiased estimates of population proportions, the statistics we examined. None of the alternatives that use unweighted model parameter estimates are effective in removing biases, even when the propensity of volunteering is correctly modeled. In contrast, when propensities are estimated using reference survey weights, estimation weights for the volunteer survey can be constructed that yield nearly unbiased estimates of proportions. The estimation weights can be either inverses of individual propensities or the inverse of average propensities within subclasses.

One of our main findings is that propensity poststratification, which is one of the methods that has been recommended in the literature on Web surveys, leads to extremely biased estimates when used alone. In propensity poststratification, weights are assigned to the volunteer sample to make its distribution equal to that of the reference survey cases among subclasses formed from estimated propensity scores. Biases result regardless of whether reference survey weights are used to estimate parameters in the propensity model. Consequently, we recommend that this estimator never be used. The defects of this method may have been obscured in previous work because propensity poststratification was combined with raking or poststratification to other more effective covariates.

When volunteering does not depend on analysis variables collected in the volunteer survey, omitting the propensity adjustment step and calibrating the volunteer sample directly to population totals of the covariates via a generalized regression estimator can also produce approximately unbiased estimates. The requirement for this to be true is that analysis variables are reasonably well described by a linear model that involves known covariates. As illustrated by Dever et al. (2008), the set of covariates may have to be fairly extensive to remove the bias. The fact that the regression estimator is competitive with any of the propensity estimators in our simulations raises the

question of whether the expense of conducting a reference survey is worthwhile. Some vendors make the argument that having a reference survey allows additional “lifestyle” or “webographic” covariates to be collected in both the reference and volunteer surveys. The hope is that these additional covariates can be used to help remove selection bias in the volunteer survey. Published empirical support for this claim is limited. Schonlau et al. (2007) found, in an example using a telephone reference survey, that phone and propensity-adjusted Web survey estimates were significantly different for a number of characteristics, but this difference was largely eliminated if demographic and webographic covariates were used to estimate propensities. On the other hand, Lee (2006) found that adding webographics was ineffective in her application. How much, if at all, a reference survey can improve estimation requires further study.

If the probability of volunteering depends on analysis variables whose means or totals are to be estimated from the survey, then none of the alternatives studied here gives unbiased estimates. This is similar to the problems caused by nonignorable nonresponse in a probability sample but seems qualitatively worse because the selection of the initial volunteer panel is not under the sampler’s control. This nonignorability is likely the greatest threat to inference using volunteer samples. There are a variety of motivations for participating in a volunteer panel—making your opinions known, being affiliated with a group, or collecting money or merchandise when incentives are paid are a few. All of these may be related to items collected in a survey and lead to the kinds of biases we have illustrated.

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Notes

1. Because the expanded Michigan Behavioral Risk Factor Surveillance Survey (BRFSS) is used as the target population, the BRFSS sample design and associated analysis weights are ignored for the simulation study.
2. Define the logistic probability of unit k 's volunteering as p_k ($k = 1, \dots, N$ with $N = 50,000$). In Poisson sampling, each unit is independently given a probability p_k of being a volunteer. The expected number of sample volunteers is $E(n) = \sum_1^N p_k$. We scaled the p_k 's as $p_k^* = p_k n^* / E(n)$, where n^* was the desired number of volunteers. As a result, $\sum_1^N p_k^* = n^*$.

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Bios

Richard Valliant is a research professor in the Joint Program for Survey Methodology at the Universities of Michigan and Maryland. His areas of research are sampling theory, model diagnostics for survey data, and variance estimation from complex survey data.

Jill A. Dever is a senior statistician at RTI International. Her areas of research are survey estimation with estimated control totals, calibration estimation, and sample design optimization.